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Assignment 1

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 99 = 3 + 96 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:

3 points: Correct answer with work shown

2 points: Incorrect answer but attempt shows some understanding (work shown)

1 point: Incorrect answer but an attempt was made (work shown)

0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts]

(link to your live google doc)

<https://docs.google.com/document/d/1dzAgyy7UhyrmR9CylVDsXdToMj1JATrVwGZaqPQtogo/edit?usp=sharing>

I. Categorical Data (15 pts = 5 x 3 pts)

1) Below is a list of unorganized data from the 2020 election. This gives the state, their geographical location, and the voting majority.

<i>State</i>	<i>Location</i>	<i>Electoral College Vote in 2020</i>
AL	South	Republican
AK	West	Republican
AZ	West	Democrat
AR	South	Republican
CA	West	Democrat
CO	West	Democrat
CT	Northeast	Democrat
DE	South	Democrat
FL	South	Republican
GA	South	Democrat
HI	West	Democrat
ID	West	Republican
IL	Midwest	Democrat
IN	Midwest	Republican
IA	Midwest	Republican
KS	Midwest	Republican
KY	South	Republican

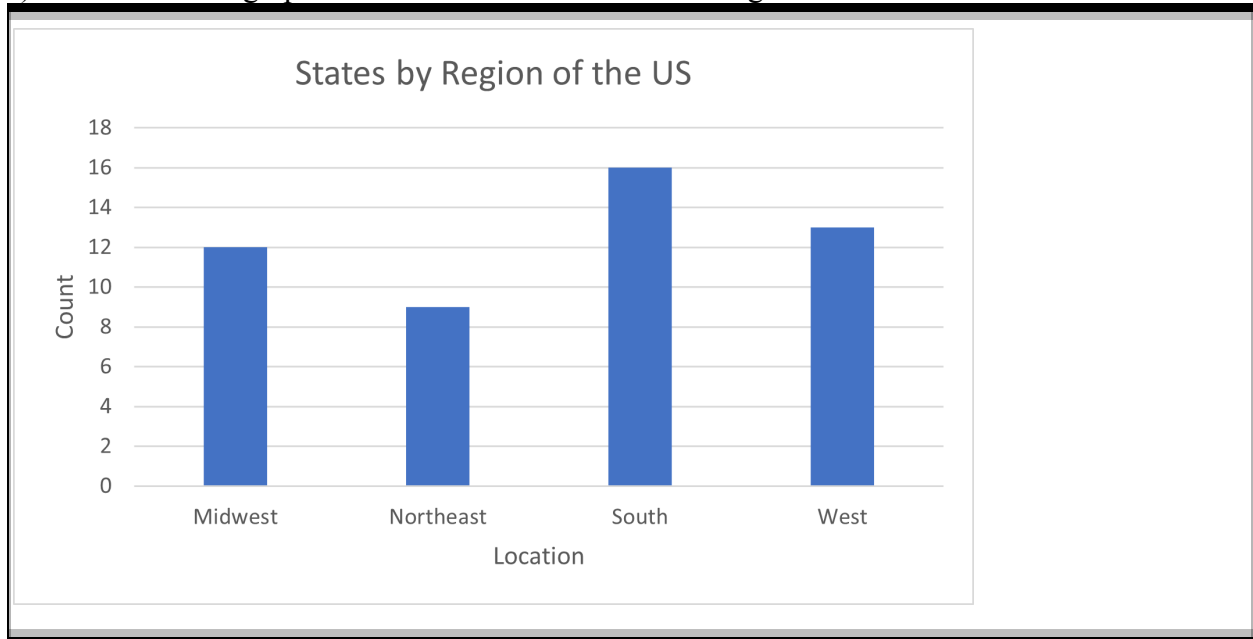
LA	South	Republican
ME	Northeast	Democrat*
MD	South	Democrat
MA	Northeast	Democrat
MI	Midwest	Democrat
MN	Midwest	Democrat
MS	South	Republican
MO	Midwest	Republican
MT	West	Republican
NE	Midwest	Republican*
NV	West	Democrat
NH	Northeast	Democrat
NJ	Northeast	Democrat
NM	West	Democrat
NY	Northeast	Democrat
NC	South	Republican
ND	Midwest	Republican
OH	Midwest	Republican
OK	South	Republican
OR	West	Democrat

PA	Northeast	Democrat
RI	Northeast	Democrat
SC	South	Republican
SD	Midwest	Republican
TN	South	Republican
TX	South	Republican
UT	West	Republican

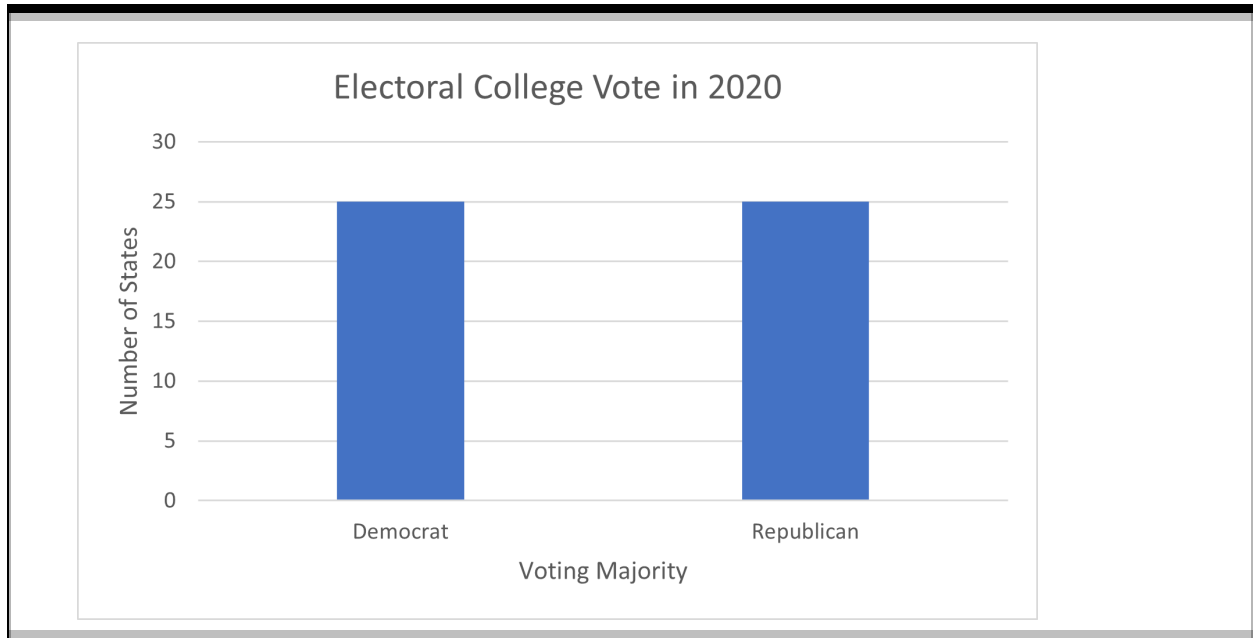
VT	Northeast	Democrat
VA	South	Democrat
WA	West	Democrat
WV	South	Republican
WI	Midwest	Democrat
WY	West	Republican

*Split Votes

a) Construct a bar graph of the amount of states in each region of the US.



b) Construct a bar graph of the amount of states who voted either republican or democrat in the 2020 election.



c) Create a two way table showing the distribution of region and political party.

Voting Majority	Location				Grand Total
	Midwest	Northeast	South	West	
Democrat	4	9	4	8	25
Republican	8		12	5	25
Grand Total	12	9	16	13	50

2) You measure the age, marital status and earned income of a random sample of 1463 women. The number and type of variables you have measured is

- (A) 1463; all quantitative.
- (B) four; two categorical and two quantitative.
- (C) four; one categorical and three quantitative.
- (D) three; two categorical and one quantitative.
- (E) three; one categorical and two quantitative.

Answer:

(E) three; one categorical and two quantitative

3) As part of a survey of college students, a researcher is interested in the variable class standing. She records a 1 if the student is a freshman, a 2 if the student is a sophomore, a 3 if the student is a junior, and a 4 if the student is a senior. The variable class standing is

- (A) Categorical
- (B) Quantitative
- (C) Quantitatively Categorical
- (D) All of the above

Answer:

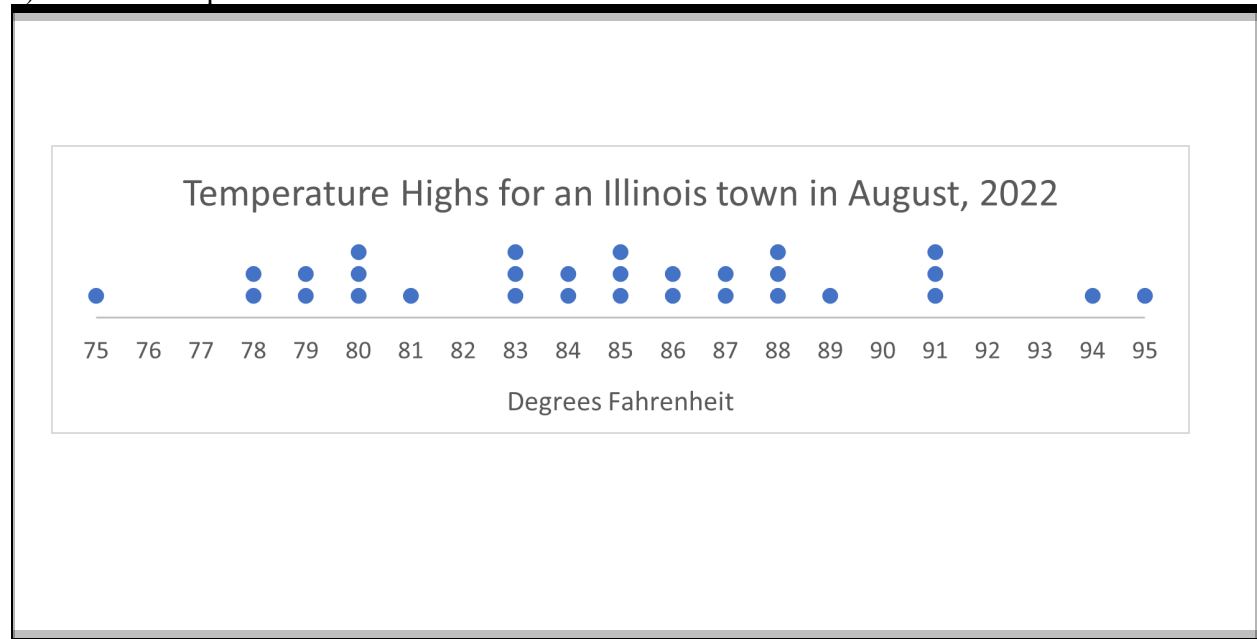
(A) Categorical

II. Quantitative Data (24 pts = 8 x 3 pts)

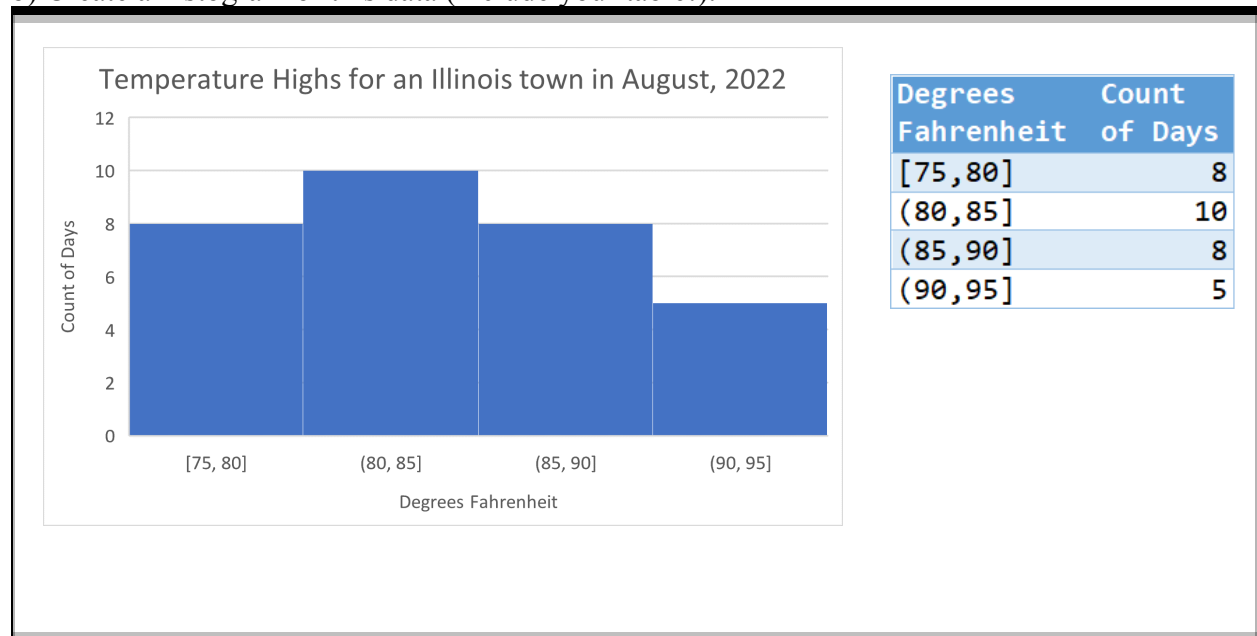
1) The following are the temperature highs for an Illinois town in August, 2022.

78	75	83	85	87	91	78	91	85	84	80	79	83
85	86	80	88	94	89	86	88	84	81	85	83	80
79	87	88	91	95								

a) Create a dot plot of this data.



b) Create a histogram of this data (include your table!).



c) Create a stem and leaf plot for this data.

Temperature Highs for an Illinois town in August, 2022

7	5 8 8 9 9
8	0 0 0 1 3 3 3 4 4
8	5 5 5 5 6 6 7 7 8 8 8 9
9	1 1 1 4
9	5

Legend

8 | 3 represents a temperature value of 83 degrees Fahrenheit

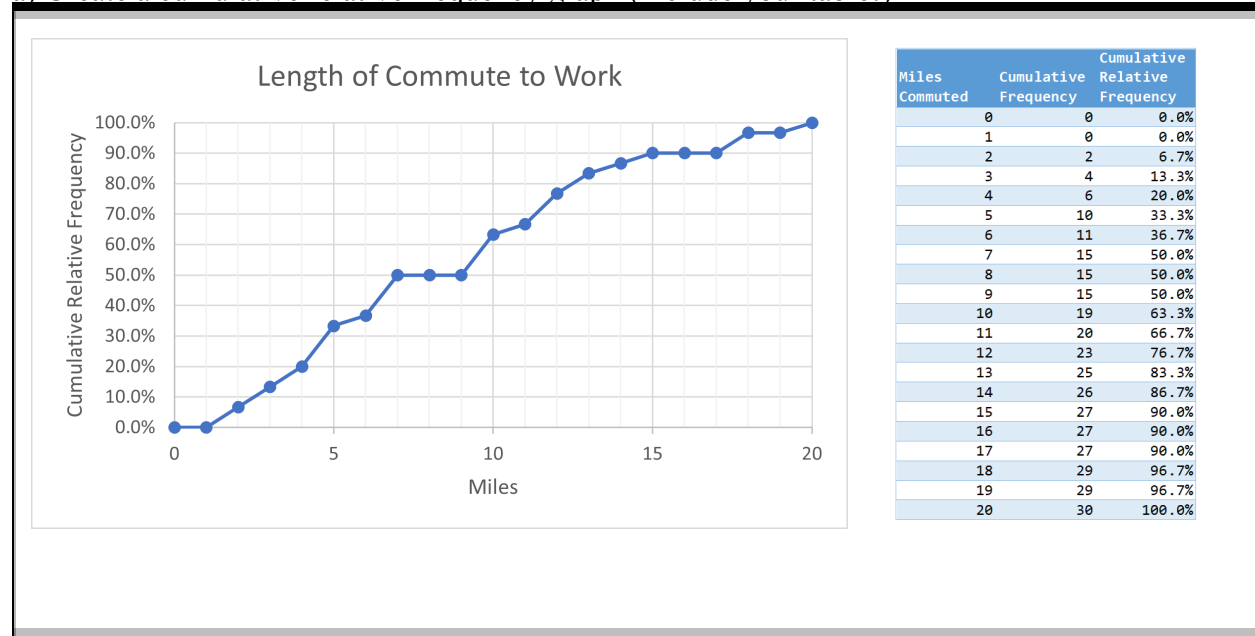
d) Which graph do you think displays the data the best? Why?

I think the dot plot displays the data the best. This is because there is a limited range of values that are possible. The temperature highs will typically be between 70 and 100 degrees in August. Therefore, the dot plot only needs to be 30 values across, which is easy to plot and visualize each value individually. The total number of data points is also limited to 31 which is easy to visualize each point individually. If the range is much greater, the data has more precision, or the number of data points is greatly increased, then a histogram would be a better option.

2) 30 people were asked how many miles, to the nearest mile, they commute to work each day:

2 5 7 3 2 10 18 15 20 7 10 18 5
 12 13 12 4 5 10 7 10 12 7 3 4 11
 13 14 5 6

a) Create a cumulative relative frequency graph (include your table!)



b) What percent of people traveled less than 12 miles?

Answer:

76.7%

c) What percent of people traveled over 15 miles?

Answer:

$100 - 90 = 10\%$

d) What percent of people traveled between 10 and 15 miles?

Answer:

$90 - 63.3 = 26.7\%$

III. Summary Statistics and Boxplots (57 pts = 19 x 3 pts)

1) Below is a list of calories and cholesterol amounts in 4 randomly selected menu items from 4 different fast food companies.

<i>Company</i>	<i>Menu Item</i>	<i>Calories</i>	<i>Cholesterol (mg)</i>
McDonald's	Bacon, Egg, & Cheese Biscuit	460	215
McDonald's	Big Mac®	590	85
McDonald's	Filet-O-Fish®	390	30
McDonald's	Cheeseburger	300	40
Burger King	Whopper® Sandwich with Cheese	740	115
Burger King	Cheeseburger	280	45
Burger King	Crispy Chicken Sandwich	670	60
Burger King	Bacon, Egg & Cheese Biscuit	400	170
Wendy's	Baconator®	960	155
Wendy's	Bacon Double Stack®	440	65
Wendy's	Classic Chicken Sandwich	490	75
Wendy's	Sausage, Egg, and Cheese Biscuit	580	285
Chick-fil-A	Chicken Biscuit	460	45
Chick-fil-A	Bacon, Egg, and Cheese Biscuit	420	180
Chick-fil-A	Grilled Chicken Sandwich	390	75
Chick-fil-A	Chicken Nuggets (8 count)	250	80

a) Using R or a graphing calculator, find the following summary statistics for CALORIES for each of the companies. (12 pts = 4 x 3 pts)

	Mean	Min	Q1	Med	Q3	Max	Std. Dev
McDonald's	435	300	367.5	425	492.5	590	122.34
Burger King	522.5	280	370	535	687.5	740	218.23
Wendy's	617.5	440	477.5	535	675	960	235.57
Chick-fil-A	300	250	355	405	430	460	91.29

Which company has the greatest calorie variability in the distribution?

Answer:

Wendy's

b) There are many methods for determining outliers. Two methods frequently used are:

Rule #1: An outlier is a value greater than $1.5 \times \text{IQR}$ above the third quartile or more than $1.5 \times \text{IQR}$ below the first quartile.

Rule #2: An outlier is a value located 2 or more standard deviations above, or below, the mean.

Using rule #1, are there any outliers in the Wendy's distribution? Show your work.

Answer:

No outliers

Q1 = 477.5
Q3 = 675
IQR = 197.5
 $1.5 \times \text{IQR} = 296.25$
Lower bound = $Q1 - 1.5 \times \text{IQR} = 181.25$
Upper bound = $Q3 + 1.5 \times \text{IQR} = 971.25$
All values are within the range

Using rule #2, are there any outliers in the Wendy's distribution? Show your work.

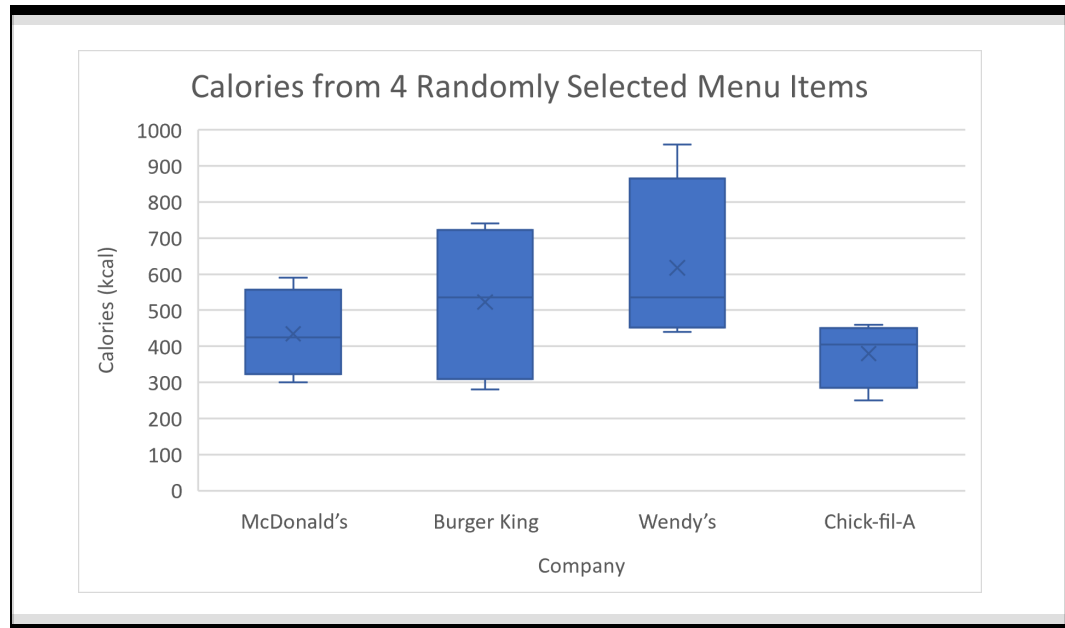
Answer:

No outliers

Mean = 617.5
Std. Dev. = 235.57
 $2 \times \text{SD} = 471.14$
Lower bound = $\text{Mean} - 2 \times \text{SD} = 146.37$
Upper bound = $\text{Mean} + 2 \times \text{SD} = 1088.63$
All values are within the range

c) Draw four modified boxplots comparing the calories for each of the four companies.

Answer:



d) In your town, McDonald's and Burger King are on the Northside and Wendy's and Chick-fil-A are on the Southside. Using R, find the following summary statistics for CHOLESTEROL for the Northside and Southside fast food restaurants. **(6 pts = 2 x 3 pts)**

	Mean	Min	Q1	Med	Q3	Max	Std. Dev
Northside	95	30	43.75	72.5	128.75	215	67.08
Southside	120	45	72.5	77.5	161.25	285	81.37

Which region has the greatest cholesterol variability in the distribution?

Answer:

Southside

e) Using rule #1, are there any outliers in the Southside's distribution? Show your work.

Answer:

No outliers

Q1 = 72.5
Q3 = 161.25
IQR = 88.75
 $1.5 * IQR = 133.125$
Lower bound = $Q1 - 1.5 * IQR = -60.625$
Upper bound = $Q3 + 1.5 * IQR = 294.375$
All values are within the range

Using rule #2, are there any outliers in the Southside's distribution? Show your work.

Answer:

Yes, 285 is an outlier

Mean = 120
Std. Dev. = 81.37
 $2 * SD = 162.74$
Lower bound = $Mean - 2 * SD = -42.74$
Upper bound = $Mean + 2 * SD = 282.74$
There is one value outside the range (285)

f) Remove the value of 285mg from the Southside cholesterol's data set. Use R to find the following values again:

	Mean	Min	Q1	Med	Q3	Max	Std. Dev
Southside	96.43	45	70	75	117.5	180	50.39

What values changed the most? What values changed the least?

**Answer
(Most):**

Max

**Answer
(Least):**

Min

_____ 2) The five-number summary for a one-variable dataset is {5, 18, 20, 40, 75}. If you wanted to construct a modified boxplot for the dataset, what would be the maximum possible length of the right side “whisker”?

- (A) 35
- (B) 33
- (C) 5
- (D) 55
- (E) 73

$$\begin{aligned} &\text{Lesser of } 1.5 * (40 - 18) \text{ or } 75 - 40 \\ &1.5 * (40 - 18) = 33 \\ &75 - 40 = 35 \end{aligned}$$

Answer:

(B) 33

_____ 3) Which of the following are true statements?

- T I. The standard deviation is the square root of the variance.
- T II. The standard deviation is zero only when all values are the same.
- T III. The standard deviation is strongly affected by outliers.

- (A) I and II
- (B) I and III
- (C) II and III
- (D) I, II, and III

(E) None of the above gives the complete set of true responses.

Answer:

(D) I, II, and III

_____ 4) In order to rate TV shows, phone surveys are sometimes used. Such a survey might record several variables, some of which are listed below. Which of these variables are categorical?

- I. The type of show being watched **Categorical**
- II. The number of persons watching the show **Quantitative**
- III. The ages of persons watching the show **Quantitative**
- IV. The name of the show being watched **Categorical**
- V. The number of times the show has been watched in the last month **Quantitative**

- (A) II, III, and V
- (B) I only
- (C) I and V
- (D) I and IV

(E) None of the above describes the complete set of correct responses

Answer:

(D) I and IV

THE END